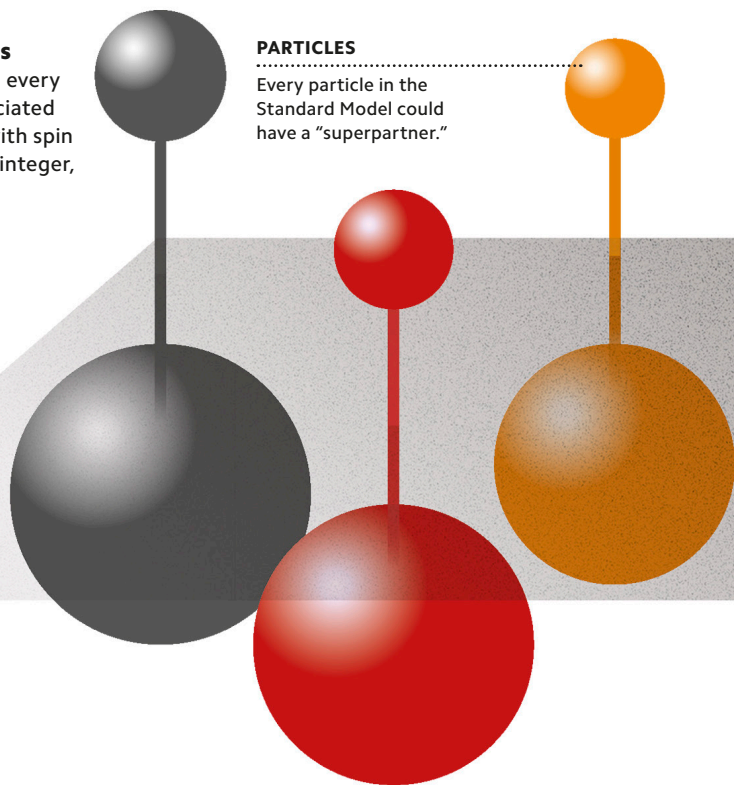


Particle partners

In supersymmetry, every boson has an associated massive fermion with spin differing by a half-integer, and vice versa.

PARTICLES

Every particle in the Standard Model could have a "superpartner."



NOT SO STANDARD

The Standard Model leaves many mysteries unresolved; for instance, it does not incorporate gravity or dark matter. Supersymmetry is a proposed extension that predicts a supersymmetric partner for every particle, with identical quantum numbers except spin, to resolve some problems with the model (such as by providing a dark matter candidate). Scientists explore beyond the Standard Model through high-energy particle accelerator experiments (see p.121).